DAY 1: Introducing Bar Graphs with “Favorite School Lunch”

Subject: Math

Essential Question(s):

- What is a bar graph?
- How can we organize data in a bar graph?
- How can we read a bar graph?

Goals/Objectives: We will be able to

- Understand what a bar graph is.
- Understand the parts of a bar graph (two axes, labels, title, frequency axis starting at zero).
- Understand how to represent data in a bar graph.
- Read the bar graph and answer questions.

Standards:

- **CCSS.Math.Content.2.MD.D.10** Draw a picture graph and a bar graph (with single-unit scale) to represent a data set with up to four categories. Solve simple put-together, take-apart, and compare problems using information presented in a bar graph.
- **MP.4** Model with mathematics.
- **MP.6** Attend to precision.

Assessment Strategies:

- Listen for students’ reasoning during discussions
- Observe student in-class work
- Students’ bar graph worksheet
- Exit slip

Connections with Prior Learning/Assets:

- In 1st grade, students learned to make and read pictographs to show simple data and compare “more” or “less”.
- This lesson builds on students’ knowledge of pictographs by introducing bar graphs as another visual tool to show categorical data.
- Students bring real-life experiences of comparing preferences (“Which is your favorite lunch?”) and can connect their reasoning to familiar contexts.

Elements of Universal Design for Learning:

- Engagement
  - Familiar, personally relevant topic (“favorite school lunch”).
  - Hands-on graph creation
  - Partner talk
  - Whole-group discussion
- Representation
  - Visual display of data.
  - Visual modeling of the graph.
  - Color-coded bar drawing with grid lines.
  - Use of relatable data (school lunches) to make meaning concrete.
- Action & Expression
  - Students can write or draw during the exit slip.
  - Oral, visual, and written participation are all valid ways to demonstrate understanding.

Materials Needed:

- Projector
- “School lunch” data
- Student worksheet (record table + bar graph)
- Colored pencils, crayons, or markers for shading bars
- Anchor chart for recording class “Noticings about Bar Graphs”
- Exit slip

Differentiation/Scaffolding:

- Bold words
- Visuals
- Gestures and pointing while naming graph parts to support comprehension
- Color-coded bar sections for visual clarity
- Model how to do the first one together
- Check-ins with specific learners who might need extra support
- Differentiated worksheet available

	• Sentence stems available • Allow students to choose the order of categories in their own tables to promote ownership • Early Finisher Task **Vocabulary/Key Words:** • Data • Category • Frequency • Axis/Axes • Bar graph • Title • Compare

Lesson Sequence - Day 1

- **Ask:** What is your favorite school lunch?
- **Say:** I asked the same question to many of my friends, and here are their answers.
    - Show the data in a table on screen.

Pizza	Tacos	Pasta	Tacos	Pizza	Tacos
Tacos	Pizza	Tacos	Burgers	Pasta	Pizza
Pasta	Tacos	Pizza	Pizza	Tacos	Tacos
Tacos	Pizza	Pasta	Tacos	Burgers	Pizza
Burgers	Tacos	Tacos	Pizza	Pasta	Tacos

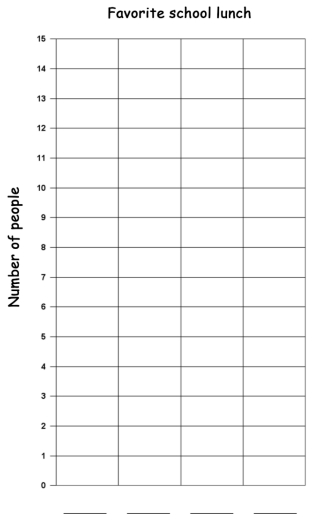
- **Ask:** What do you notice about the responses?
    - **Anticipated Student Response:** A lot of people like pizza and tacos. Not many like burgers.
- **Ask:** Which lunch seems most popular? Is it easy to tell just by looking at this table?
    - **Anticipated Student Response:** Pizza. Tacos. It's hard to tell.
- **Say:** I agree. It's hard to tell which school lunch is the most popular when the data is shown like this. How could we organize the data to make it clearer? Turn and talk to your shoulder partners to share your ideas.
    - (Student turn-and-talk, then share as a whole group)
- **Ask:** Who would like to share some ideas?
    - **Anticipated Student Response:** pictograph; bar graph; use tally marks; make a chart…
- **Say:** Those are all great ideas! Today, we'll learn how to organize our data using a bar graph. But first, let's put our data into a record table.
    - Model counting aloud and recording "9" for "Pizza"
        - Write "Pizza" in a table

School Lunch	Number
Pizza	9

- **Ask:** Which lunch should we record next?
    - Call on one student, then **Ask:** Can you count for us?
    - Record ideas
- **Say:** Now fill in the names of the four school lunches in your own tables (in any order), count, and record the total number for each.

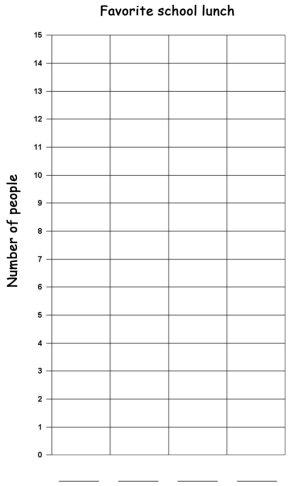
School Lunch	Number

- **Do:** Circulate around the rug to check.
- **Say:** Let's come back together. Who would like to share what they got next?
    - Invite students to the projector to help complete my table.
- **Ask:** Now that our data table is organized, what do we all notice?
    - **Anticipated Student Response:** Five chose pasta. A lot chose pizza. Tacos are the most popular school lunch.
- **Say:** Let's see how we can show this information in another way, with a bar graph.
- **Do:** Show a blank bar graph grid on screen labeled "Favorite School Lunch" (The x-axis = Lunches, y-axis = Number of people, starting at zero with single-unit scale)



- **Ask:** What do you notice about this graph?
    - **Anticipated Student Response:** It has numbers on the side. The numbers show the number of people. Each square is one person. There's space for each school lunch on the bottom. The title is "Favorite school lunch".
- **Say:** Yes! So there is a title over here, and there are two axes, the category axis over here for school lunches and the frequency axis over here for number of people. Everyone, say "axis".
    - (Students repeat)
- **Say:** And I noticed that the number axis starts at zero. Can someone tell me why?
    - **Anticipated Student Response:** Because we start counting from zero.
- **Say:** Right! Every bar graph starts at zero so we can see exactly how many there are. The names of the lunches go along the bottom. On my table, "Pizza" is first, so I'll put "Pizza" in the first box.
- **Do:** Model using the scale and grid lines to find the top of the bar (think aloud about how to count carefully to reach the correct height), draw in the bar and shade it in.
    - **Think aloud:** I recorded 9 in my table for pizza, so I need to count up to 9 to show that…(draw a stop line at 9) see how the bar stops right at 9?
    - Draw and shade in the first bar
    - **Say:** They are called bar graphs because they have bars to represent the data. Where the bar ends tells us how many there are in that category.
- **Say:** Let's try another one together. Who would like to come up and help me? Loud and proud.
    - Call on students to draw the next bar
- **Say:** Now it's your turn! Use the data in your table to complete your own bar graph.

Name: _____



- Circulate, guide, and check student work
- Early Finisher Task (Reorder categories alphabetically or from greatest to least)
- Call on students to help fill in the large class bar graph

- **Ask:** Now that we've finished our bar graphs, let's see if we can tell which school lunch is the most popular. Let's say it together.
  - **Anticipated Student Response (whole-group):** Tacos!
- **Ask:** How do you know?
  - **Anticipated Student Response:** Because it has the tallest bar.
- **Ask:** What does each bar tell us?
  - **Anticipated Student Response:** It shows how many people like that lunch.
- **Say:** Awesome! See how a bar graph makes it so much easier to see and compare data? Now look at your graph and talk with your partner: What else do you notice about the bar graph?
  - (Student discuss, then share back)
- **Closing:** Let's come back from our discussions. Who would like to share what they notice from the bar graph?
  - **Anticipated Student Response:** The graph has 4 categories. There are numbers on the side that go up by one. There are labels on the bar graph and a title./Burgers are the least favorite lunch. Two more people like pasta than burgers.
  - Record student thinking on shared chart.
- **Ask:** Nice! Before we wrap up for today, I want to see what you noticed from our work with bar graphs. I'm going to send out an exit slip to each of you. On the exit slip, there's a question that asks "What parts does every bar graph need?" followed by an empty box. You can write or draw your ideas. When you're finished, please make sure your name is on the paper and put it into the turn-in bin.
  - Circulate as they write

What do you notice about bar graphs?

(You can write or draw your ideas!)

or

What do you notice about bar graphs?

(You can write or draw your ideas!)

I notice that _____.

The tallest bar shows _____.

Each bar shows _____.

- **Say:** You all did a great job today learning how to organize data, create, and read a bar graph. Tomorrow, we will explore more about bar graphs. We'll collect our own data and make our own bar graph!

DAY 2: Creating and Interpreting a Whole-Class Bar Graph ("Favorite Season")

Subject: Math

Essential Question(s):

- How can we collect data and represent data in a bar graph?
- How can we read and understand the information shown in a bar graph?
- How can we use a bar graph to answer questions?

Goals/Objectives: We will be able to

- Engage in shared data collection.
- Understand how to represent data in a bar graph.
- Read and interpret data on a bar graph to answer questions.

Standards:

- **CCSS.Math.Content.2.MD.D.10** Draw a picture graph and a bar graph (with single-unit scale) to represent a data set with up to four categories. Solve simple put-together, take-apart, and compare problems using information presented in a bar graph.
- **MP.4** Model with mathematics.
- **MP.6** Attend to precision.

Assessment Strategies:

- Observe student participation during discussions
- Review what students wrote/drew on sticky notes
- Exit slip

Connections with Prior Learning/Assets:

- Builds directly on Lesson 1, where students learned bar-graph parts and created graphs from a given data set.
- Connects to real-life experiences (everyone has a favorite season) to make data personally relevant.

Elements of Universal Design for Learning:

- Engagement
  - Familiar, personally relevant topic ("favorite season").
  - Active voting and bar-building
  - Sticky note reflection
  - Partner talk
  - Whole-group discussion
- Representation
  - Visual display of data on board.
  - Visual modeling of the graph on chart paper.
  - Color-coded bar drawing with grid lines.
  - Concrete connection to student experiences (seasons).
- Action & Expression
  - Students can write or draw during the exit slip.
  - Oral, visual, and written participation are all valid ways to demonstrate understanding.

Materials Needed:

- Projector
- Bar graph from Lesson 1
- Sticky notes
- Colored pencils, crayons, or markers for shading bars
- Chart paper

Differentiation/Scaffolding:

- Bold words
- Visuals
- Gestures and pointing to support comprehension
- Color-coded bar sections for visual clarity
- Model how to do the first one together

● White board and markers ● Exit slip	● Check-ins with specific learners who might need extra support ● Sentence stems available ● Early Finisher Task **Vocabulary/Key Words:** ● Data ● Category ● Frequency ● Axis/Axes ● Bar graph ● Title ● Compare ● Most/Fewest ● More/Less ● Together

Lesson Sequence - Day 2

- **Ask:** Yesterday we learned how to organize data in a bar graph and the parts of a bar graph. What do you remember about the parts of bar graphs?
    - **Anticipated Student Response:** It has two axes. It has labels. It has a title. The frequency axis starts at zero.
    - Scaffolding (if needed): Point to yesterday's bar graph to prompt recall.
- **Say:** Perfect! Today, we are going to collect and organize real data from our own class and create our class bar graph together on this chart paper. Let's start by collecting data. Think about your favorite season. What's your favorite season? Put your heads down and raise your hand when you hear your favorite season.
- **Say:** Fall… Winter… Spring… Summer.
    - Count hands and record the numbers in a table on the board

Season	Number

- **Say:** All right, everyone, look up. Here's our class data table showing how many students chose each season. Let's organize the information into a bar graph. Who can tell me what should the title of our bar graph be?
    - **Anticipated Student Response:** Favorite season.
    - Write it on the chart paper.
- **Ask:** Great! What should our labels be?
    - **Anticipated Student Response:** Seasons, Number of students
    - Label the x- and y-axes as students answer
- **Ask:** On the frequency axis, what number should I start with?
    - **Anticipated Student Response:** Zero.
- **Say:** Exactly. Bar graphs always start at zero so we can see the full amount.
- **Do:** Begin with one category (e.g., Winter).
- **Ask:** How high should I go for Winter?
    - (Students call out ideas)
    - Draw a "stopper line" at that number, then shade to make the bar.
- **Do:** Continue the same process for the remaining seasons.
    - Invite a few students to come up and draw bars with guidance.
- **Say:** Now that our bar graph is complete, take a quiet moment to look at it and think about what you notice. When you have something to share, please don't shout it out yet, show me a quiet thumbs-up on your chest so that I know

you are ready.

- **(Once most students are ready) Say:** Turn and talk to your neighbor and share what you noticed about the bar graph.
    - Circulate to listen to student discussions.
- **Say:** Let's come back together. I'm passing out a sticky note to each of you. On it, write or draw one thing you noticed about our bar graph. Make sure your name is on it. When you are done, raise your hand and I'll collect it and stick it around our graph.
- **Do:** Circulate, collect notes, and post them around the graph.
    - Sentence stems (if needed): I notice that ___ has more than ___. The bar for ___ is taller than ___.
- (Early Finisher: give more sticky notes)
- **Say:** Look at all these great noticings! Let's read a few together. If you agree or noticed the same thing, show me (hand gesture). Who would like to start this off?
    - Invite students to share aloud and use the graph to justify answers.
- **Closing:** Wonderful sharing! You did a great job reading our class bar graph! Now I have an exit slip for you. It has a few questions about the graph we just created.
    - Distribute exit slips

Name: _____

Exit Slip

(You can write or draw your ideas)

1. How many students chose fall as their favorite season?

2. Which season did the fewest students choose?

3. How many students chose spring and summer together?

    - Scaffold (if needed): Read questions aloud
- **Say:** I feel so proud of how you interpreted data on the bar graph today. Tomorrow, we will work in tables to collect your own data and make your own bar graphs about something you choose! And then we will read each other's graphs and answer questions about them.

DAY 3: Creating and Interpreting Our Own Bar Graphs

Subject: Math	
Essential Question(s): ● How can we organize and represent data in a bar graph? ● How can we use a bar graph to solve problems?	
Goals/Objectives: We will be able to ● Collect our own data. ● Organize and represent our data in a bar graph. ● Read and interpret data on a bar graph to solve simple put-together, take-apart, and compare problems. **Standards:** ● **CCSS.Math.Content.2.MD.D.10** Draw a picture graph and a bar graph (with single-unit scale) to represent a data set with up to four categories. Solve simple put-together, take-apart, and compare	**Assessment Strategies:** ● Observe student participation during group work and discussions ● Listen for mathematical reasoning during group work ● Review group data record table and bar graphs ● Question sheets ● Exit slip

problems using information presented in a bar graph. ● **MP.4** Model with mathematics. ● **MP.6** Attend to precision.	

Connections with Prior Learning/Assets:
● Builds on Lesson 1 (understanding bar-graph parts) and Lesson 2 (shared class graph creation and interpretation).
● Activates prior skills in collecting and interpreting categorical data.
● Encourages cooperative decision-making and communication
● Leverages students' cultural and linguistic assets through topic choice, where they can select data meaningful to their interests and lived experiences (e.g., favorite food, game, pet).
	Elements of Universal Design for Learning:
	● Engagement
	  ○ Student-chosen topics.
	  ○ Collaborative graph creation
	  ○ Whole-group discussion
	● Representation
	  ○ Visual models on projector to review steps.
	  ○ Clear examples of record tables and bar graphs.
	  ○ Explicit cues for structure (title, labels, axes).
	● Action & Expression
	  ○ Students can write or draw during the exit slip.
	  ○ Oral, visual, and written participation are all valid ways to demonstrate understanding.

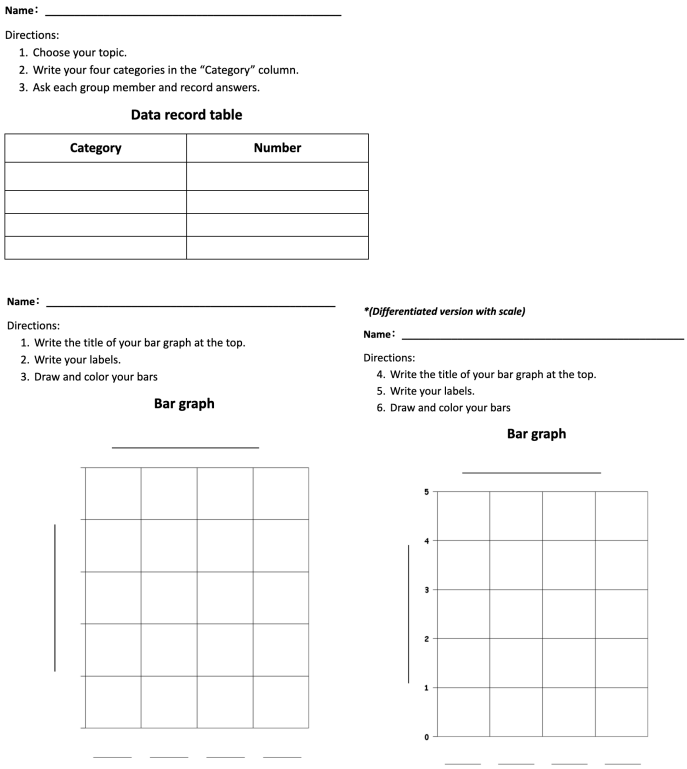
Materials Needed: ● Projector ● Group data-collection record table ● Group bar-graph templates (blank grids) ● Question sheet templates ● Colored pencils, crayons, or markers for shading bars ● White board and markers ● Exit slip	**Differentiation/Scaffolding:** ● Bold words ● Visuals ● Gestures and pointing to support comprehension ● Check-ins with specific learners who might need extra support ● Sentence stems available ● Early Finisher Task **Vocabulary/Key Words:** ● Data ● Category ● Frequency ● Axis/Axes ● Bar graph ● Title ● Compare ● Most/Fewest ● More/Less ● Put together ● Take apart

Lesson Sequence - Day 3

- **Say:** Yesterday we made a bar graph for our favorite season. Do you still remember which season was the most popular?
  - Invite quick responses.
- **Ask:** What else do we know about bar graphs so far?
  - **Anticipated Student Response:** They have two axes. They start at zero. They have labels and titles.
- **Say:** Awesome! Today, we are going to work in our table groups and create our own bar graphs from start to finish. Here's what you'll do: First, pick a topic together as a group, something you are all curious about, like "Favorite Fruit", "Favorite Pet", or "Favorite Recess Game". Next, you need to collect data by asking each group member and record

the information in your record table. Finally, show your data in the bar graph. Don't forget your titles and labels so everyone knows what your graph is showing! Also, you do not need to fill in all four categories, but you must have at least two. Can someone remind us: How many categories do we need at least?

- Show worksheet on projector and point to each section while explaining.



Name: _____

Directions:
1. Choose your topic.
2. Write your four categories in the "Category" column.
3. Ask each group member and record answers.

Data record table

Category	Number

Name: _____

Directions:
1. Write the title of your bar graph at the top.
2. Write your labels.
3. Draw and color your bars

Bar graph

(Differentiated version with scale)

Name: _____

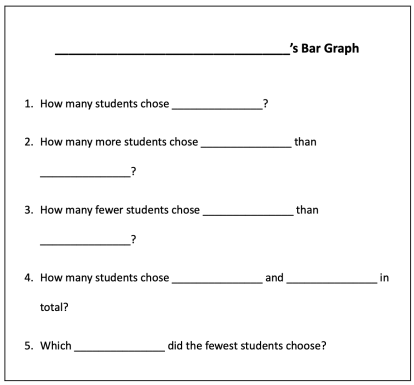
Directions:
4. Write the title of your bar graph at the top.
5. Write your labels.
6. Draw and color your bars

Bar graph

- Write on board: "1. Choose topic 2. Collect and record data 3. Create bar graph *Include title and labels!"

- **Say:** You may go back to your tables now. Work together to pick your topic and start collecting data. Make sure everyone's name is written on your group worksheet.
  - Send students back by table #.
- **Do:** Circulate and ask probing questions (How do you decide your categories?)/offer support.
  - Have an extra eye on Table #1 and #4 (which consist of 6 students) to ensure every student contributes (and not too chatty)
- **Say:** When your group finishes the graph, bring it to me so I can check it. Then, I'll give you a question sheet. You will design and fill in 5 questions for another group to answer about your graph.

Question Sheet



_____'s Bar Graph

1. How many students chose _____?
2. How many more students chose _____ than _____?
3. How many fewer students chose _____ than _____?
4. How many students chose _____ and _____ in total?
5. Which _____ did the fewest students choose?

- (Once every group is finished)
- **Say:** Now trade your completed graph and question sheet with another group. When you get a new graph, take a moment to read it carefully before answering the questions. Use the bars to help you solve.
  - Trade graphs and question sheets
  - Hand out paper to write down answers to each student

Name: _____

1. _____

2. _____

3. _____

4. _____

5. _____

- **Do:** Circulate to listen and support groups as they discuss and solve (prompting Qs: How do you know? How did you find that total?).
- **Say:** Please put both your graph and question sheets into the turn-in bin and return to your rug spot.
- **Ask:** How do you like creating your own bar graph and reading others' graphs?
- **Ask:** What is one thing you would like to share from another group's graph?
- **Closing:** Those were awesome findings! Over these lessons, we've learned a lot about bar graphs. We've learned the parts of a bar graph, how to record data, and how to use bar graphs to solve problems. Before we end today, I want you to think about this question: "What did you learn about how bar graphs help us?" I'm going to send you the exit slips. Put it in the turn-in bin once you are finished.
    - Distribute exit slips.

Exit Slip

- What did you learn about how bar graphs help us?

  One thing I learned about bar graphs is _____

  _____.

    - Circulate while students write or draw their ideas.
- **Say:** Nice work today, everyone! You collected data, created graphs, and interpreted each other's graphs, just like real data scientists! You can go back to your seats.